

Experts Problem and Online Convex Optimization

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References

- Lecture 1 and 2 of [Haipeng's course](#)

Learning from Experts (The Experts Problem)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss of every expert $\ell_t = (\ell_t(1), \dots, \ell_t(N)) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle = \mathbb{E}_{i \sim p_t} [\ell_t(i)]$

A special case of Online Convex Optimization with

$\mathcal{X} = \Delta_N$ and $f_t(p) = \langle p, \ell_t \rangle$

$$\text{Regret} := \sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=1}^T \ell_t(i)$$

Follow the Leader (FTL)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

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Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

Define
$$L_t(i) := \sum_{s=1}^t \ell_s(i)$$

“Follow the leader” strategy: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$ and let $p_t = e_{i_t}$

Follow the Leader (FTL)

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

$$\text{Regret} = \underbrace{\sum_{t=1}^T \langle p_t, \ell_t \rangle}_{\geq T-1} - \underbrace{\min_{i \in [N]} \sum_{t=1}^T \ell_t(i)}_{\approx \frac{T}{2}} \approx (T-1) - \frac{T}{2} \approx \underline{\Theta\left(\frac{T}{2}\right)} \neq o(T)$$

Softmin

Replacing “min” by “softmin”:

$$p_t(i) = \frac{w_t(i)}{\sum_{j=1}^N w_t(j)} \quad \text{where} \quad w_t(i) = e^{-\eta L_{t-1}(i)} \quad \textbf{(Exponential weights)}$$

“Follow-the-leader” (hardmin) is a special case when $\eta = \infty$

Regret Guarantee for Exponential Weights

Theorem 1.

Assume that $\eta \ell_t(i) \geq -1$ for all t, i . Then Exponential Weight ensures

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) \ell_t(i)^2$$

Assume $0 \leq \ell_t(i) \leq 1$

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta T = 2\sqrt{T \log N} \leq o(T)$$

$$\eta = \sqrt{\frac{\log N}{T}}$$

$$\log = \ln$$

Proof of Theorem 1

Recall $L_t(i) = \sum_{s=1}^t l_s(i)$

and $P_t(i) = \frac{w_t(i)}{W_t}$ where $w_t(i) = \exp(-\eta L_{t-1}(i))$, $W_t = \sum_{j=1}^N w_t(j)$

$$\begin{aligned} \ln \frac{W_{t+1}}{W_t} &= \ln \frac{\sum_{i=1}^N w_{t+1}(i)}{W_t} = \ln \frac{\sum_{i=1}^N \exp(-\eta L_t(i))}{W_t} \\ &= \ln \frac{\sum_{i=1}^N \underbrace{w_t(i)}_{\exp(-\eta L_{t-1}(i))} \exp(-\eta l_t(i))}{W_t} \\ &= \ln \frac{\sum_{i=1}^N w_t(i) \exp(-\eta l_t(i))}{W_t} = \ln \left(\sum_{i=1}^N P_t(i) \exp(-\eta l_t(i)) \right) \end{aligned}$$

$$(L_t(i) = L_{t-1}(i) + l_t(i))$$

$$\ln \frac{W_{t+1}}{W_t} = \ln \left(\sum_{i=1}^N p_t(i) \exp(-\eta l_t(i)) \right)$$

$$\leq \ln \left(\sum_{i=1}^N p_t(i) (1 - \eta l_t(i) + \eta^2 l_t(i)^2) \right)$$

$$= \ln \left(\sum_{i=1}^N p_t(i) - \eta \sum_{i=1}^N p_t(i) l_t(i) + \eta^2 \sum_{i=1}^N p_t(i) l_t(i)^2 \right)$$

$$= \ln \left(1 - \eta \langle p_t, l_t \rangle + \eta^2 \sum_{i=1}^N p_t(i) l_t(i)^2 \right)$$

$$\leq -\eta \langle p_t, l_t \rangle + \eta^2 \sum_{i=1}^N p_t(i) l_t(i)^2$$

For real number $z \leq 1$

$$\exp(z) \leq 1 + z + z^2$$

$$z = -\eta l_t(i)$$

$$\ln(1+z) \leq z$$

Summing over $t=1, 2, \dots, T$

$$\ln \frac{W_{T+1}}{W_1} \leq -\eta \sum_{t=1}^T \langle p_t, l_t \rangle + \eta^2 \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

$$\boxed{\ln \frac{W_{T+1}}{W_1}} \leq -\gamma \sum_{t=1}^T \langle p_t, l_t \rangle + \gamma^2 \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

$$i^* = \operatorname{argmin}_i L_T(i)$$

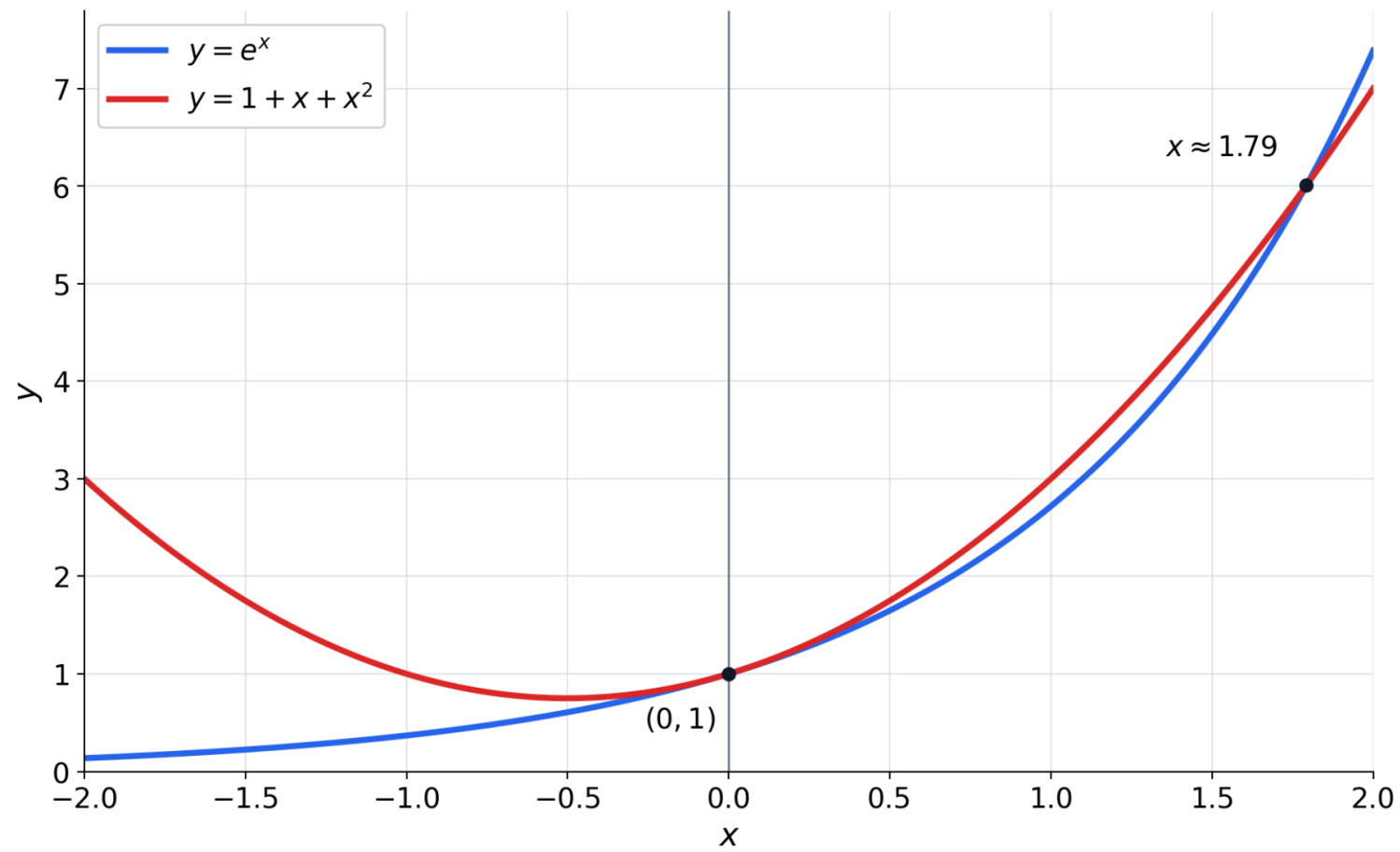
Recall $W_t = \sum_{i=1}^N \exp(-\gamma L_{t-1}(i))$

$$\begin{cases} W_1 = \sum_{i=1}^N \exp(0) = N \\ W_{T+1} = \sum_{i=1}^N \exp(-\gamma L_T(i)) \geq \exp(-\gamma L_T(i^*)) \end{cases} \quad \underline{L_0(i) = 0}$$

$$\Rightarrow \ln \frac{W_{T+1}}{W_1} = \ln W_{T+1} - \ln W_1 \geq -\gamma L_T(i^*) - \ln N$$

$$\Rightarrow -\gamma L_T(i^*) - \ln N \leq -\gamma \sum_{t=1}^T \langle p_t, l_t \rangle + \gamma^2 \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

$$\Rightarrow \sum_{t=1}^T \langle p_t, l_t \rangle - \underbrace{L_T(i^*)}_{\text{min}} \leq \frac{\ln N}{\gamma} + \gamma \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$



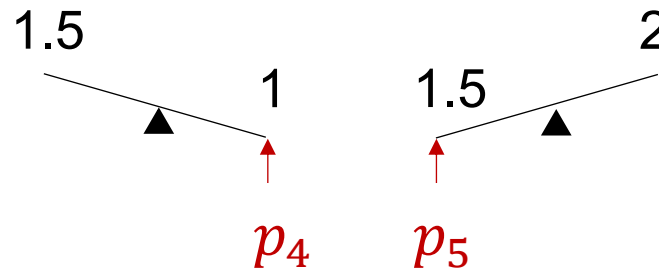
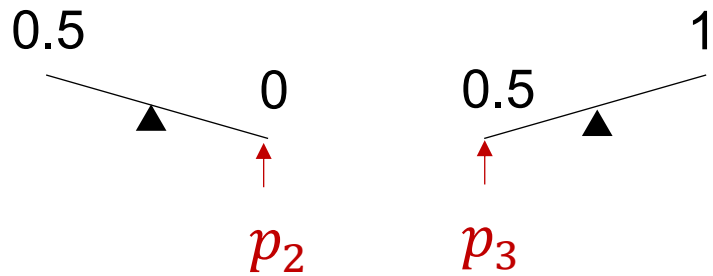
Why softmin works, while min does not?

(Discussion)

The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

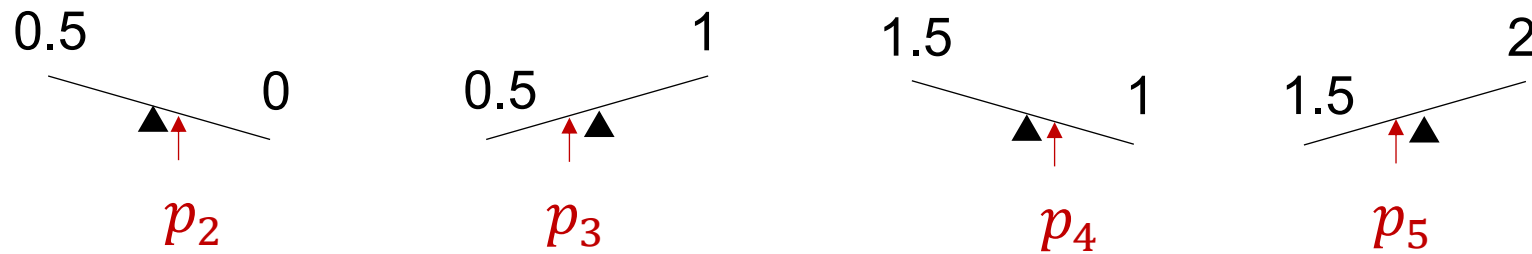
FTL:
$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$$



The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

EW: $p_t(i) \propto e^{-\eta L_{t-1}(i)}$



Why Stability Helps?

(Discussion)

$$\text{dist}_{\text{ave}}(P_t, P_{t+1})$$

$\nearrow$
 l_t

Be-The-Leader Lemma

Theorem 2. Follow the leader (FTL) strategy

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \sum_{s=1}^{t-1} f_s(x)$$

ensures

$$\sum_{t=1}^T f_t(x_t) - \min_x \sum_{t=1}^T f_t(x) = \text{Regret} \leq \sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1}))$$

If we have stability:

$$x_t \approx x_{t+1}$$

(This lemma doesn't need convexity of f_t or $\mathcal{X}$)

$$\sum_{t=1}^T f_t(x_{t+1}) - \min_x \sum_{t=1}^T f_t(x) \leq 0$$

Proof of Theorem 2

$$\text{By FTL: } \chi_{t+1} = \arg \min_x \sum_{s=1}^t f_s(x)$$

Define $\phi_t = \min_x \sum_{s=1}^t f_s(x) = \sum_{s=1}^t f_s(\chi_{t+1})$

$$\phi_{t-1} = \min_x \sum_{s=1}^{t-1} f_s(x) = \sum_{s=1}^{t-1} f_s(\chi_t)$$

$$\leq \sum_{s=1}^{t-1} f_s(\chi_{t+1})$$

($\because \chi_t$ is the minimizer)

$$= \sum_{s=1}^t f_s(\chi_{t+1}) - f_t(\chi_{t+1})$$

$$= \phi_t - f_t(\chi_{t+1})$$

$$\phi_T = \min_x \sum_{t=1}^T f_t(x)$$

$$\phi_0 = 0$$

$$f_t(\chi_{t+1}) \leq \phi_t - \phi_{t-1}$$

$$\sum_{t=1}^T f_t(\chi_{t+1}) \leq \sum_{t=1}^T (\phi_t - \phi_{t-1})$$

$$\leq \sum_{t=1}^T (\phi_t - \phi_{t-1})$$

$$= \phi_T - \phi_0$$

$$\sum_{t=1}^T f_t(\chi_{t+1}) \leq \min_x \sum_{t=1}^T f_t(x)$$

$$= (\phi_1 - \phi_0) + (\phi_2 - \phi_1) + \dots + (\phi_T - \phi_{T-1})$$

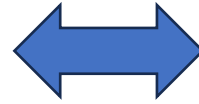
Follow The Regularized Leader

(Generalizing the Exponential Weights algorithm)

An Equivalent Form of EW

$$\psi(p) := \sum_{i=1}^N p(i) \ln p(i) \quad (\text{convex})$$

$$p_t(i) = \frac{e^{-\eta L_{t-1}(i)}}{\sum_{j=1}^N e^{-\eta L_{t-1}(j)}}$$

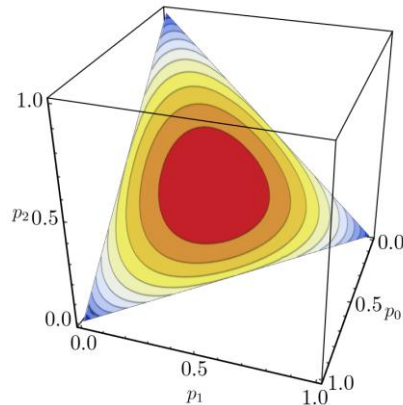


Verify yourself
(Lagrange multiplier)

$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

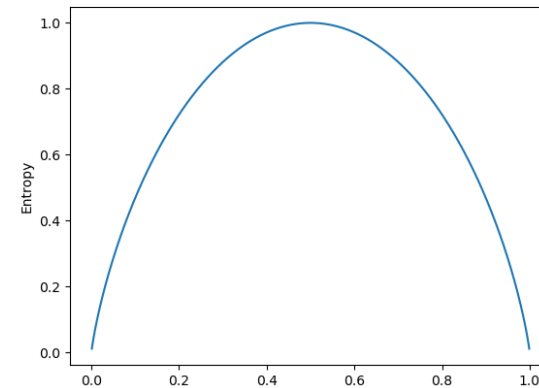
negative entropy

$-\psi(p) = \sum_{i=1}^N p(i) \ln \frac{1}{p(i)}$ is called the **(Shannon) entropy function** (concave)



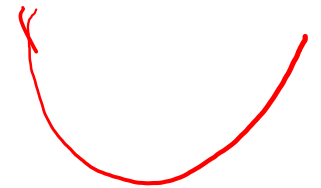
N=3

[\(link\)](#)



N=2

[\(link\)](#)



Follow the Regularized Leader (FTRL) for OCO

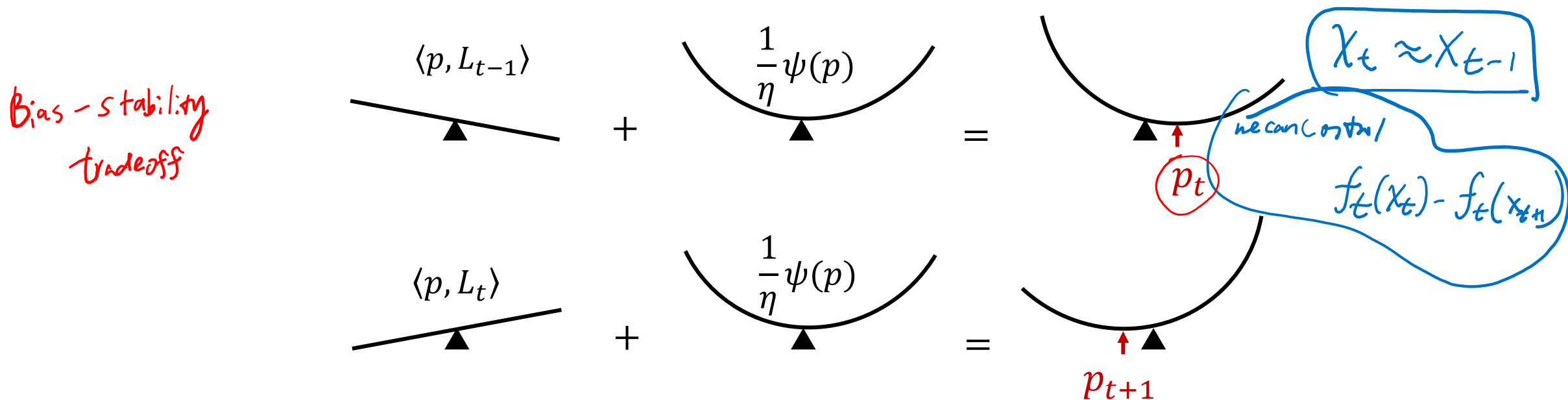
$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} \psi(x) \right\}$$

Regularizer ψ

$$\operatorname{argmin}_{x \in \mathcal{X}} \sum_{s=0}^{t-1} f_s(x)$$

define $f_0(x) = \frac{1}{\eta} \psi(x)$

If ψ is *curved*, it makes the update more stable:



Strongly Convex Functions

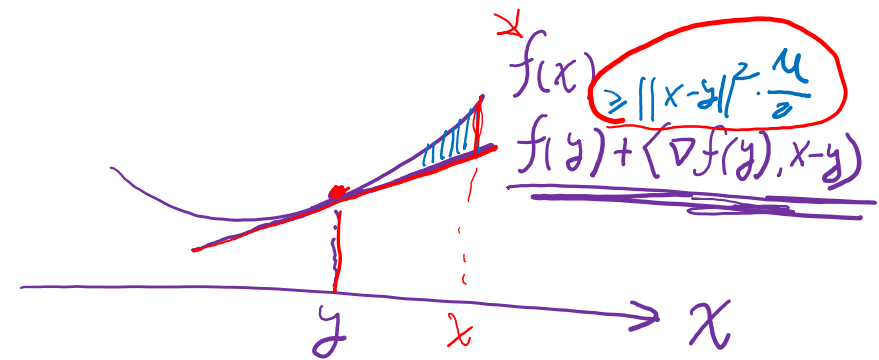
Def (convex function):

A differentiable function $\psi: X \rightarrow \mathbb{R}$ is convex if $\forall x, y \in X$

$$\psi(x) \geq \psi(y) + \langle \nabla \psi(y), x - y \rangle$$

Def (μ -strongly convex fun) ψ is called μ -strongly-convex with respect to norm $\|\cdot\|$ if $\forall x, y$

$$\underline{\psi(x)} \geq \psi(y) + \langle \nabla \psi(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2$$



Regret Guarantee of FTRL in OLO

$$\begin{aligned}\|x\|_2 &= \sqrt{x_1^2 + \dots + x_d^2} \\ \|x\|_\infty &= \max_i |x_i| \\ \|x\|_1 &= |x_1| + |x_2| + \dots + |x_d|\end{aligned}$$

Theorem 3. Let ψ be μ -strongly-convex in norm $\|\cdot\|$.

Then FTRL in OCO ensures,

$$\sum_{t=1}^T f_t(x_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T f_t(x) \leq \frac{\max_{u \in \mathcal{X}} \psi(u) - \min_{u \in \mathcal{X}} \psi(u)}{\eta} + \frac{\eta}{\mu} \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2$$

$\|\cdot\|_*$ is the dual norm of $\|\cdot\|$

$$\begin{aligned}\frac{O(1)}{\eta} + O(2) \sum_{t=1}^T G^2 &= O\left(\frac{1}{\eta} + 2TG^2\right) \\ \eta &\approx \frac{1}{G\sqrt{T}} \quad O(G\sqrt{T})\end{aligned}$$

Zinkevich. Online Convex Programming and Generalized Infinitesimal Gradient Ascent. 2003.

Zinkevich. Online Convex Programming and Generalized Infinitesimal Gradient Ascent. 2003.

Proof of Theorem 3 (Stability)

By BTL lemma,

$$\forall x, \quad \sum_{t=0}^T f_t(x_t) - \sum_{t=0}^T f_t(x) \leq \sum_{t=0}^T (f_t(x_t) - f_t(x_{t+1}))$$

$$\Rightarrow \sum_{t=0}^T f_t(x_{t+1}) \leq \sum_{t=0}^T f_t(x)$$

$$\Rightarrow \frac{1}{\eta} \psi(x_1) + \sum_{t=1}^T f_t(x_{t+1}) \leq \frac{1}{\eta} \psi(x) + \sum_{t=1}^T f_t(x)$$

$$\Rightarrow \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x) \leq \underbrace{\frac{1}{\eta} \psi(x) - \frac{1}{\eta} \psi(x_1)}_{\text{Bias } (\approx \frac{1}{\eta})} + \underbrace{\sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1}))}_{\text{Stability } (\approx \eta)}$$

$$x_t = \underset{x}{\operatorname{argmin}} \left\{ \sum_{s=0}^{t-1} f_s(x) \right\}$$

$$f_0(x) = \frac{1}{\eta} \psi(x)$$

$$\langle p_t, l_t \rangle - \langle p_{t+1}, l_t \rangle$$

Given $\|\cdot\|$, $\|x\|_* \triangleq \sup_{\|y\| \leq 1} \langle x, y \rangle$

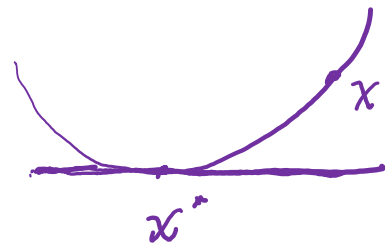
$$\langle a, b \rangle \leq \|a\| \|b\|_*$$

Proof of Theorem 3 (Stability)

Goal: bound $f_t(x_t) - f_t(x_{t+1}) \leq \underbrace{\langle \nabla f_t(x_t), x_t - x_{t+1} \rangle}_{\substack{\uparrow \\ \text{by convexity of } f_t}} \leq \boxed{\|x_t - x_{t+1}\|} \boxed{\|\nabla f_t(x_t)\|_*} \leq \frac{\eta}{2} \|\nabla f_t(x_t)\|_*^2$

Lemma: Let $F: \mathcal{X} \rightarrow \mathbb{R}$ be a μ -strongly convex function, and let $x^* = \arg \min_{x \in \mathcal{X}} F(x)$. Then for any $x \in \mathcal{X}$,

$$F(x) \geq F(x^*) + \frac{\mu}{2} \|x - x^*\|^2$$



Proof: By strong convexity, $F(x) \geq F(x^*) + \langle \nabla F(x^*), x - x^* \rangle + \frac{\mu}{2} \|x - x^*\|^2$

By first-order optimality $\langle \nabla F(x^*), x - x^* \rangle \geq 0$ for any x

Proof of Theorem 3

$$x_t = \operatorname{argmin}_{x \in X} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} \psi(x) \right\}$$

$$F_t(x)$$

F_t is $\frac{\mu}{\eta}$ -strongly convex

$$F_{t+1}(x) - F_t(x) = f_t(x)$$

By the previous lemma :

$$\begin{cases} F_t(x_{t+1}) \geq F_t(x_t) + \frac{\mu}{2\eta} \|x_t - x_{t+1}\|^2 \\ F_{t+1}(x_t) \geq F_{t+1}(x_{t+1}) + \frac{\mu}{2\eta} \|x_t - x_{t+1}\|^2 \end{cases}$$

$$F_t(x_{t+1}) - F_{t+1}(x_{t+1}) + F_{t+1}(x_t) - F_t(x_t) \geq \frac{\mu}{\eta} \|x_t - x_{t+1}\|^2$$

$$\Rightarrow \frac{\mu}{\eta} \|x_t - x_{t+1}\|^2 \leq \underbrace{f_t(x_t) - f_t(x_{t+1})}_{\leq \|x_t - x_{t+1}\| \| \nabla f_t(x_t) \|_*}$$

$$\Rightarrow \|x_t - x_{t+1}\| \leq \frac{\eta}{\mu} \| \nabla f_t(x_t) \|_*$$

Special Case: Negative Shannon Entropy

$$\mathcal{X} = \Delta_N, \quad \psi(x) = \sum_{i=1}^N x(i) \log x(i)$$

(Pinsker's ineq)

ψ is 1-strongly concave w.r.t. $\|\cdot\|_1$

$$\psi(x) \in [-\log N, 0]$$

$$f_t(x) = (x, l_t)$$

$$\nabla f_t(x) = l_t$$

$$\text{Regret} \leq \frac{\ln N}{2} + 2 \sum_{t=1}^T \|l_t\|_\infty^2$$

(max_i l_t(i)²)

previous

$$\sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

Special Case: Squared Euclidean Norm

$\mathcal{X}$ = general convex set, $\psi(x) = \frac{1}{2} \|x\|_2^2$

ψ is 1-strongly convex w.r.t. $\|\cdot\|_2$

Assume $\|x\|_2 \leq R \quad \forall x \in \mathcal{X}$

$$R_{\text{regret}} \leq \frac{\frac{1}{2} R^2}{\gamma} + \gamma \sum_{t=1}^T \|\nabla f_t(x_t)\|_2^2$$

Remark: The Role of Entropy Regularization

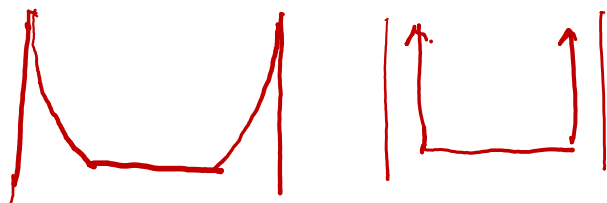
Exploration: $p_t(i) > 0 \quad \forall i$

(ψ has steep boundary)

Stability: $p_t \approx p_{t+1}$

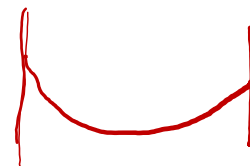
(ψ has curvature) ← property we use here

Exploratory but not stable:



ξ -greedy

Stable but not exploratory:



$$\psi(x) = \frac{1}{2} \|x\|_2^2$$

OCO with only gradient feedback

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Learner makes a decision $x_t \in \mathcal{X}$

Adversary decides a convex loss function $f_t: \mathcal{X} \rightarrow \mathbb{R}$

Adversary reveals $\nabla f_t(x_t)$ to the learner

Online Mirror Descent

Online Mirror Descent

Implicit Online Mirror Descent

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ f_t(x) + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

Online Mirror Descent

(Problem 3 of Homework 1)

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

The regret guarantee and analysis are closely related to those of FTRL

Follow The Perturbed Leader

Online Linear Optimization with a Weaker Adversary

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Adversary decides a **linear** loss function $f_t(x) = \langle x, \ell_t \rangle$
(not revealing it to the learner at this moment)

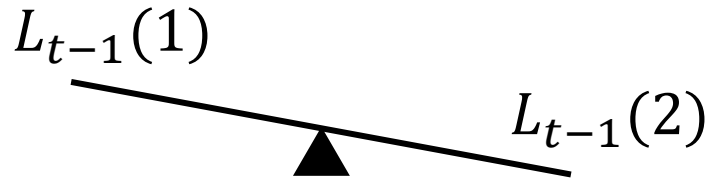
Learner makes a decision $x_t \in \mathcal{X}$

Adversary reveals ℓ_t to the learner



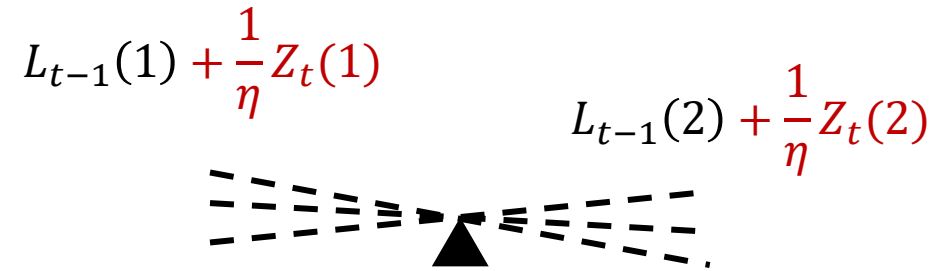
order swapped

Another Way to Avoid Exploitation by Adversary



$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$$

Follow the Leader



For example: $Z_t(i) \sim \mathcal{N}(0, \sigma^2) \quad \forall i = 1, 2$

$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i) + \frac{1}{\eta} Z_t(i)$$

Follow the Perturbed Leader

Similar to FTRL, “randomization” does not precisely capture the key mechanism of FTPL. The key is “stability.”

FTPL for OLO

Follow The Perturbed Leader (FTPL) for OLO

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} \langle x, \ell_s \rangle + \frac{1}{\eta} \langle x, Z_t \rangle \right\} = \operatorname{argmin}_{x \in \mathcal{X}} \left\langle x, L_{t-1} + \frac{1}{\eta} Z_t \right\rangle$$

where $Z_t \in \mathbb{R}^d$ is drawn from some distribution $\mathcal{D}$.

When is FTPL more preferred than FTRL?

For many common decision set $\mathcal{X}$, it is (computationally) much easier to solve

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle$$

than

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle + \psi(x)$$

for a given $L \in \mathbb{R}^d$.

When is FTPL more preferred than FTRL?

Example: Online Shortest Path

Given: a graph, a source, and a sink

Let $\mathcal{Y} \subset \{0,1\}^{|E|}$ be the set of paths from the source to sink

For $t = 1, 2, \dots, T$:

Adversary decides the costs on all edges $\ell_t \in \mathbb{R}^{|E|}$

Learner chooses a path $y_t \in \mathcal{Y}$

Adversary reveals ℓ_t to the learner

$\mathcal{Q}$ above is not a convex set, but we can

- Run OCO in $\mathcal{X} = \text{convhull}(\mathcal{Y})$ (becoming a set of flows)
- Sample y_t such that $\mathbb{E}[y_t] = x_t$ in round t .

“Smooth” Perturbations

Regret Guarantee of FTPL in OLO

Theorem 4. Let $\mathcal{D}$ be a κ -smooth distribution.

Then FTPL in OLO ensures,

$$\sum_{t=1}^T \langle x_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x_t, \ell_t \rangle \leq \frac{\mathbb{E}_{Z \sim \mathcal{D}} \left[\max_{u \in \mathcal{X}} \langle u, Z \rangle \right] - \mathbb{E}_{Z \sim \mathcal{D}} \left[\min_{u \in \mathcal{X}} \langle u, Z \rangle \right]}{\eta} + \eta \kappa \sum_{t=1}^T \|\ell_t\|_*^2$$

(Problem 4 of HW1)

Key Step in the Regret Analysis